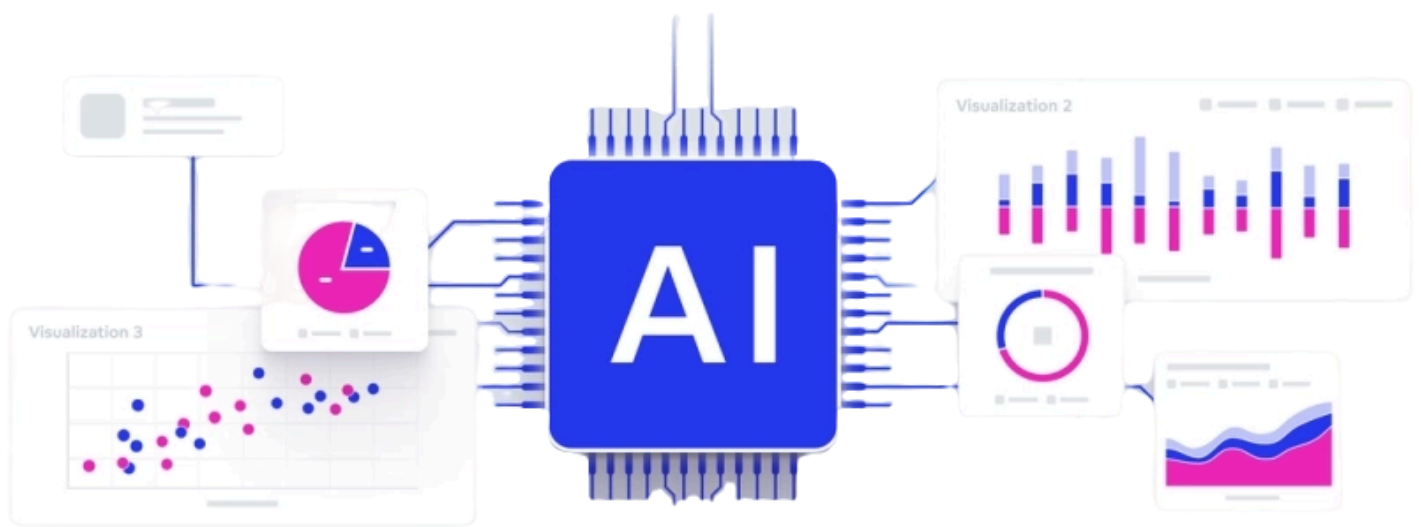


ENTRI elevate

In-Demand Coding Program



DIGITAL HANDBOOK

AI-DRIVEN DATA ANALYTICS

AI-Driven Data Analytics program is a technology training program that will be of high industry relevance in the coming years. By going through the program, students will be able to achieve certification that will help them to apply for different job roles in many Technology and Business organizations (such as Information Technology, Banking, Financial Services, Retail) that apply such technologies.



SYLLABUS

1. Data Analysis Using Excel

UNIT 1. Introduction to Data Analytics, Data Analytics Lifecycle, Types of Data Statistics Basics (Mean, Median, Mode, Variance, Standard Deviation). Introduction to Excel Interface, Basics of Excel, Spreadsheet basics, Data Entry. Fundamentals of Excel, Insertion, Deletion, Importing Data, Table Creation Introduction to Excel Formulas - sum(), min(), max(), count(), average(), if(), sumif(), countif(), averageif(), formatting text using right(), left(), mid(), upper(), lower(), proper().

UNIT 2. Power Query Introduction, Introduction to Data Cleaning, Removing duplicate data, Handling missing data, Correcting inconsistent data, Text-to-columns and Flash Fill, Splitting and merging data, Data Validations, Data transformation, What-if analysis, scenario Manager, Removing unnecessary characters and spaces, Conditional formatting for data cleaning.

UNIT 3. Data exploration and summary statistics, Filtering and Sorting data in excel, Descriptive Statistics, Mean, standard deviation, etc., using Analysis ToolPak, Mean, standard deviation, etc., using Analysis ToolPak. Date and Time Functions: TODAY, NOW, YEAR, MONTH, WEEKDAY, DATEDIF, Lookup and Reference: VLOOKUP, HLOOKUP, XLOOKUP, INDEX, MATCH

UNIT 4. Introduction to Pivot tables, Creating Pivot tables, Connecting multiple tables and show analysing and creating visualization with multiple tables. Data Analysis and Visualization using Pivot Table.

Data Visualization: Charts and Graphs, Bar, Column, Pie, Line, Scatter, Combo charts, Sparklines, PivotCharts, Dynamic charts with slicers and filters, create and design interactive dashboard.

AI Integration: Quick Analysis for charts & summaries, AI analyst for formulas & insights

2. PowerBI

UNIT 1. Understanding the Power BI Ecosystem, Installation of PowerBI Desktop, Exploring the Power BI interface, Connecting to data sources. Introduction to Power Query, Importing Data into Power Query, Data Cleaning and Transformation: Handling Missing Data and Duplicate Data, Sorting and Filtering Rows, Aggregating and Grouping Data in Power Query.

Scorecard

We will be maintaining a Scorecard to keep track of your assessment scores -- assignments, module-end assignments, projects, and final exams. This scorecard will be issued to the eligible candidates at the end of the course along with the certificate. Check **Page 5** for the Eligibility Criteria.



SYLLABUS

2. PowerBI

UNIT 2. Introduction to Data Modeling, Normalization vs denormalization, Relationships in PowerBI, Cardinality, cross filter direction, Building Star schema data model, What is DAX? , Calculated Columns and tables using DAX, Measures, Simple DAX Functions and Operators (Mathematical, Logical, Statistical & Text Functions), Advanced DAX – Aggregations and Calculations (Aggregate, Filter, Datetime, Time Intelligence & Table Functions)

UNIT 3. Data Visualization & Formatting - Bar, Column, Pie, Donut, Line, Area, Scatter, KPI & Gauge, Table, Matrix, Map, Tree map, Combo Applying Conditional Formatting And Data Bars ; Creating Hierarchies ; Filters, Slicers ; Drill-Downs Creating a comprehensive report in PowerBI, Create bookmarks and buttons, Design a great PowerBI dashboard and publishing into PowerBi service.

AI Integration: Analyze via right-click, built-in AI visuals like Q&A and Key Influencers for insights from natural language queries.

MINI PROJECT

Intermediary Assessment

Mid-Level Certification

3. MySQL

UNIT 1. Introduction to Databases, Overview of RDBMS, Normalization, Overview of MySQL Overview of MySQL Workbench, Installation of MySQL Workbench, Table and Terminologies. Overview of SQL Commands (DDL,DML, DCL, TCL) and Datatypes. DDL - CREATE, ALTER, RENAME, TRUNCATE, DROP Keys- PK, FK, SK, CK, composite, non-prime attribute. Constraints - unique, not null, check, default, auto_increment ; DML - INSERT, SELECT, UPDATE, DELETE, DROP vs TRUNCATE vs DELETE.

UNIT 2. SQL Clauses - DISTINCT, AS, WHERE, Operators (Arithmetic, Comparison, Logical). Other Operators - IS NULL, IS NOT NULL, IN, NOT IN, BETWEEN, NOT BETWEEN, LIKE, NOT LIKE, ORDER BY, LIMIT, Aggregate functions, GROUP BY, HAVING, INNER JOIN, LEFT JOIN, RIGHT JOIN, UNION, UNION ALL.

UNIT 3. Built-in function: Mathematical, String, Date & time; User-defined functions, Stored Procedures, Subqueries - single, multi-row; Views - simple, complex views, Triggers, DCL- GRANT, REVOKE; TCL - COMMIT, SAVEPOINT, ROLLBACK - ACID properties, show connecting MySQL and PowerBI

AI Integration: Text-to-SQL tools for converting English to SQL queries

What is inside Scorecard?

Formative Assessments: These assessments occur during the learning phase of the course to monitor learner progress with ongoing feedback. This includes the weekly assignments.

Comprehensive Assessments: These assessments occur during the learning phase of the course to assess the learners overall understanding of each module and their skills. This includes the module end projects.

Summative Assessments: These assessments occur at the end of the learning process to evaluate the knowledge and skills of the learner. This includes the MCQ and Codeathon.



SYLLABUS

4. Python Fundamentals

UNIT 1. Introduction to Programming Languages and Python, Installing Anaconda and Jupyter Notebook. Google Colab walkthrough, Basic Python Syntax and Data Types. Data Structures - String, String methods, String slicing, Tuple, Tuple methods, Immutable vs Mutable Data Types, Dictionary, Dictionary methods, Set, Set methods, Frozenset.

UNIT 2. List creation, List methods, List slicing, Operators; Conditional statements - if, else, elif, while loop; else, break & continue with while loop, range; for loop; else, break & continue with for loop; nested for loop; enumerate; List Comprehension. Build-in Functions & User-Defined Functions- Types of arguments, Local and Global Namespace. Introduction to OOPS Concepts.

5. Python for Data Analytics

UNIT 1. NumPy - NumPy arrays, Array operations and functions. Pandas - Series & Dataframe, Data Ingestion. Slicing, indexing (with index, boolean, loc, iloc) and manipulating (add, update, delete).

Data Exploration (readcsv, head, tail, shape, columns, info, dtypes, describe, value_counts, isnull, dropna, fillna). Data Wrangling - merge, concat, join, group by.

5. Python for Data Analytics

UNIT 2. Introduction to Data visualisation in Python- Visualization Using Matplotlib- Scatter Plot, Line Chart, Bar Chart, Pie Chart, Histogram, Box Plot; Visualization Using Seaborn- Count Plot, Bar plot, Scatter plot, Line plot, Box plot, Histogram, Density Plot, Violin Plot, Swarm Plot, Heatmap, Pair Plot.

UNIT 3. Intro to Plotly, Dynamic Visualization using Plotly. Data Cleaning using Pandas and Conducting Statistical Analysis.

Introduction to Webscraping, Making HTTP request to fetch data, Using BeautifulSoup to Extract data from website.

AI Integration: Gemini in Google Colab for auto-generating and explaining Python code from natural language

6. Capstone Project

You are expected to create a working project using the knowledge & skills that you have earned through this course. This project needs to be completed within 30 days where Project topic will be provided by your project mentor.

Summative Assessment | Final Course Completion Certification

Summative & Intermediary Assessment

1. You will have a maximum of 2 attempts for the Summative Assessment
2. The first attempt is free of cost, and the second attempt will be upon a cost of INR 500/-
3. Intermediary Assessment - you will have only 1 chance, that too as per the schedule

The Entri Way

★ BEST PRACTICES!

- Allocate a minimum of 12 hours per week for learning and practice
- Watch & practice all in-app videos for the current week before attending live sessions
- Join the live sessions on time and stay till the end
- Notify the course coordinator and mentor 24 hours in advance if you'll miss a session.
- Complete midweek tasks and module-end assignments on time
- Have a dedicated, distraction-free workspace for your practice and learning
- Use a 64-bit machine with a minimum of Windows 8.1 or OS X 10.
- Possess a headset with a microphone and a webcam for live sessions.

ESCALATION MATRIX

You may connect with the below members for any concerns:

- 1.Course Coordinator: Queries related to your Fees, Attendance, In-App support
- 2.Batch Mentor: Academic doubts & support
- 3.Lead Mentor: If you cannot connect with the course coordinator or batch mentor
- 4.Placement Associate: Regarding placement queries



CODE OF CONDUCT

Respect Others: We value diversity and inclusion. Treat every member of our community with respect and kindness, and refrain from any behavior that is discriminatory, harassing, or abusive.

Be Honest: Maintain academic integrity by avoiding cheating or plagiarism. Take responsibility for your work and give proper credit when using external sources.

Be Punctual: Attend all classes and be on time. If you cannot make a class or will be late, notify your instructor in advance to minimize any disruptions.

Collaborate: Embrace teamwork and cooperation. Engage with your fellow students and use Discord, our official communication platform, for doubts and helping others.

Be Communicative: Keep your microphone and video accessible during classes to foster an engaging and interactive learning environment.

Report Concerns: If you encounter behavior that violates our Code of Conduct, promptly report it to your instructor following our Escalation Matrix.

ELIGIBILITY

CERTIFICATE ELIGIBILITY

Learner can earn the course certificate only after meeting the below criteria:

1. Learner should have minimum 85% attendance for the live sessions
2. Learner should complete all the assignments, assessments and project as per the course agenda on time
3. Learners should score 40% or above separately for the Formative, Comprehensive & Summative Assessment

PLACEMENT ELIGIBILITY

- Watch all video sessions on the app
- Maintain a minimum of 85% attendance in live sessions
- Complete all capstone and module-end assignments within the scope and time
- Meet the formative and comprehensive assessment
- Successfully complete the summative assessment
- Attend and pass mock interviews



GENERAL GUIDELINES



- You are expected to adhere to the program's timeline.
- According to our teaching model, you learn the concept from the In-app videos and learn the practical implementation in the Live session.
- It is mandatory to join the session through a laptop or a desktop computer. Ideal machine requirements: 64-bit Machine, Windows 8.1 above, OS X 10.
- A headset with a microphone and webcam is essential for virtual learning, facilitating visual communication in live sessions.
- Use your official full name as the Display Name and maintain a professional display picture when interacting with mentors on all platforms.
- If unable to attend a session, notify the course coordinator and mentor at least 24 hours in advance.
- Mentors won't repeat missed sessions; review VODs and recorded live sessions independently. Post doubts on the Discord channel if needed.
- We use Discord as our official communication platform and you will be required to use it for any communication outside of live sessions. Connect with the mentors strictly through Discord only and not any other media.
- You are supposed to react or respond to mentors' messages such as polls, intimations, and updates duly, and should ask for clarity if required.
- You should be able to unmute and speak when a question is directed to you from mentors.
- Join and stay for the entire live session for attendance to count. Attendance and in-app activity will be monitored for placement eligibility.
- A reliable internet connection (10 Mbps upload and download speed) is crucial for online classes. Download the necessary software, and communicate with mentors and fellow students.
- Mock interviews are mandatory for interview preparation. Notify the coordinator 24 hours in advance if unable to attend. Missing two mock interviews will result in exclusion from placements.





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Entri Software Private Limited

Krf Business Centre, 37/2188,
Maleppally Road, Thrikkakara,
Kakkanad, Cochin, Ernakulam,
Kerala. Pin-682021



support@entri.me